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An examination of the construct validity and factor structure of the Groton Maze Learning Test, a new measure of spatial working memory, learning efficiency, and error monitoring

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Abstract

This study examined the construct validity of the Groton Maze Learning Test (GMLT) in assessing processing speed, working memory, and aspects of executive function in healthy adults. Performance on GMLT outcome measures was compared to performance on tests of psychomotor speed, working memory, and learning from the CogState computerized cognitive test battery (CGS; <http://www.cogstate.com/>). The factor structure of the GMLT was evaluated using exploratory factor analysis. The stability of this factor structure was examined in a large sample of patients undergoing parathyroidectomy or thyroidectomy. Results of this study suggest that the GMLT measures of spatial learning efficiency and error monitoring correlate with CogState measures of attention, working memory, and learning. Exploratory factor analysis yielded a two-factor solution of error monitoring and learning efficiency, which was stable across repeated assessments.

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The Groton Maze Learning Test (GMLT) is a recently developed, computer-based neuropsychological measure of immediate- and short-term memory for visuospatial information (Snyder, Bednar, Cromer, & Maruff, 2005). It is also sensitive to the detection of subclinical perseverative errors and slowed information-processing speed in healthy adults (Pietrzak, Cohen, & Snyder, 2007; Snyder, Bednar, et al., 2005; Snyder, Werth, et al., 2005). The GMLT is regularly administered as part of the CogState neuropsychological test battery (<http://www.cogstate.com/>) as a measure of spatial learning and error monitoring. It is included as part of an early phase battery, which is used in phase I studies of novel pharmaceutical and nutraceutical compounds, as well as schizophrenia, attention-deficit/hyperactivity disorder

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(ADHD), and other modifiable test batteries, which are employed by academic researchers and several major global pharmaceutical companies to serve as cognitive endpoints in clinical trials.

While several studies have examined the utility of the GMLT in assessing spatial working memory and error monitoring in clinical populations (e.g., Roman et al., 2005; Schroder, Snyder, Sielski, & Mayes, 2004; Snyder, Bednar, et al., 2005; Snyder, Werth, et al., 2005; Snyder, Maruff, Pietrzak, Cromer, & Snyder, 2007; Snyder et al., *in press*), little is known about the construct validity of the GMLT in assessing cognitive functions that underlie spatial working memory task performance (e.g., processing speed, planning, working memory, error monitoring) compared to traditional neuropsychological instruments.

Spatial working memory relies on multiple cognitive processes, including attention, visuomotor processing speed and integration, and decision-making (Darby & Walsh, 2005). Thus, tasks that assess spatial working memory may also assess related cognitive functions such as visuomotor processing speed and various aspects of executive function. For example, navigating a hidden maze such as the GMLT requires coordination of visuomotor and visuo-perceptual functions and executive processes that encode and manipulate spatial information relevant to completing the maze and that guide performance and error monitoring (Pietrzak et al., 2007; Snyder et al., *in press*). Poor performance on the GMLT may therefore reflect disruption in any of these cognitive functions or their integration. Because the GMLT yields a number of primary and secondary measures (see Table 1), it may provide information regarding cognitive operations that underlie spatial working memory. For example, scores on the GMLT-timed chase test may serve as an index of simple visuomotor processing speed, as this measure assesses the total number of moves an individual makes while tracking a stimulus through a grid in a fixed period of time. Total time, total moves, and maze efficiency index measures may serve as indices of spatial working memory, as they are sensitive to an individual's ability to temporarily store and effectively utilize information about a hidden maze from one trial to the next. Perseverative and rule-break errors may serve as an indices of performance and error monitoring, as these measures are sensitive to one's ability to follow rules, correctly respond to feedback, and inhibit pre-potent responding during this spatial working memory task. An evaluation of the convergent validity of the GMLT compared to performance on a separate battery of computerized neuropsychological tasks that assesses a variety of cognitive functions may provide a richer and more

Table 1
Primary and secondary outcome measures of the Groton Maze Learning Test

GMLT measure	Metric	Definition	Number of trials on which performance measure is based
Primary outcome measures			
Timed chase test (TCT)	Correct moves in 30 s	Number of tiles correctly touched while chasing a target through the 10 × 10 matrix, over a 30-s interval	One trial
Total time	Log10 s	Total time required to learn the pathway on each of the five learning trials in each assessment then normalized using a logarithmic base 10 transformation	Summed over five trials <i>Note:</i> If inspection of raw data distributions suggests that transformation is not necessary, then raw values are used
Total moves	Number	Total moves required to find the pathway on each of the five learning trials in each assessment.	Summed over five trials
Maze efficiency index	Mean correct moves per second	Ratio of number of correct moves (summed over five trials) to total time (normalized using a logarithmic base 10 transformation if time is non-normally distributed)	Summed over five trials, and then divided by 5 to obtain a mean value
Secondary outcome measures			
Legal errors	Number	Number of errors made whilst adhering to the three standard GMLT rules	Summed over five trials
Perseverative errors	Number	Number of times a legal error is made repeatedly (on consecutive trials)	Summed over five trials
Rule-break errors	Number	Number of errors made in contravention of GMLT standard rules (moving backwards, moving diagonally, making repeated perseverative errors)	Summed over five trials

thorough understanding of the cognitive processes involved in spatial working memory and related aspects of cognition assessed by the GMLT.

Some demonstrations of the convergent (Boulanger et al., 2006; Lee et al., 2005) and discriminant validity (Snyder et al., *in press*) of the GMLT are available. The GMLT has been compared to one measure of spatial working memory (Lee et al., 2005), but the comparator task in that study is not well known and the purpose of that experiment was not to validate the GMLT. Results of this study showed that the GMLT measure of total time required to complete five learning trials correlated well with performance on a virtual city navigation task in healthy older adults. Pietrzak et al. (2007) demonstrated moderate convergent validity for the GMLT compared to a well-known measure of fluid reasoning and problem solving (Tower of Toronto Test) and of concentration, divided attentional controls and information-processing speed (Paced Auditory Serial Addition Test), but this remains a single demonstration of this aspect of test validity. It is crucial that assessment measures first be thoroughly validated before their widespread use in research and clinical practice. Thus, a more comprehensive analysis of the convergent validity of the constructs assessed by the GMLT is required. A comparison of the GMLT to more widely used and previously validated computerized measures of processing speed, working memory, and others aspects of executive function will help determine the convergent validity of this novel hidden maze learning test.

The CogState (CGS) neuropsychological test battery consists of computerized cognitive tasks that are adaptations of standard neuropsychological tests, which assess a range of cognitive functions, including psychomotor speed, attention, working memory, and learning. This battery requires approximately 20 min to complete, and it consists of tasks in the form of card games that are presented in succession on a green background. In order to aid individuals in attaining optimal performance quickly with each task, an instructional phase presents written instructions and visual feedback to interactively teach the rules for each task. Once an individual successfully completes a sufficient number of practice trials to demonstrate their awareness of these rules, the testing phase of each task begins and instructional aids are hidden. The CogState battery contains measures of psychomotor function (simple detection task), visual attention (identification, simple matching and monitoring tasks), executive function (working memory and complex matching tasks) and memory (learning; see Section 1 for detailed descriptions of these tests).

The CogState battery was designed specifically for repeated assessment with minimal practice effects (Collie, Darby, Falletti, Silbert, & Maruff, 2002; Falletti, Maruff, Collie, & Darby, 2006). It is standardized, has a short administration time, contains multiple alternate forms, and yields data appropriate for detecting cognitive change in within-subject designs. More than 50 peer-reviewed publications have examined the clinical utility and psychometric properties of the CogState test battery, which has demonstrated validity and reliability in assessing cognitive function in ADHD (Mollica, Maruff, & Vance, 2004), mild cognitive impairment (Darby, Maruff, Collie, & McStephen, 2002; Maruff et al., 2004), concussion (Collie et al., 2004; Moriarity et al., 2004), fatigue and alcohol use (Cairney, Clough, Jaraqba, & Maruff, 2007; Falletti, Maruff, Collie, Darby, & McStephen, 2003), coronary artery bypass grafting (Silbert et al., 2004), and cognitive effects of various medications (e.g., Collie, Maruff, Snyder, Darekar, & Huggins, 2006; Maruff et al., 2006). A comprehensive list of publications and conference presentations of studies employing the CogState test battery is available at the CogState website (<http://www.cogstate.com/>).

The CogState battery is emerging as an industry standard for use in phase I and II pharmaceutical clinical trials. A number of pharmaceutical companies, including Abbott, AstraZeneca, Bristol-Myers Squibb, GlaxoSmithKline, Johnson & Johnson, and Pfizer Inc., as well as the Centers for Disease Control and Prevention and a large number of university-based clinical research laboratories, are currently using the CogState battery to monitor cognitive changes associated with the administration of novel pharmaceutical and nutraceutical products that impact central nervous system functioning.

CogState tests are an ideal choice to examine the construct validity of the GMLT. First, these tests assess cognitive functions purportedly assessed by the GMLT (e.g., processing speed, attention, and working memory), thereby providing ideal comparator measures for a study of the convergent validity of the GMLT. They also provide an opportunity to examine the divergent validity of the GMLT as a test of executive function (specifically, performance and error monitoring), as the CogState tests primarily assess attention, information-processing speed, and motor function/reaction time, but not higher-order cognitive operations such as spatial working memory and error monitoring. Second, CogState has acquired from Pfizer Inc., the rights to provide the GMLT as a commercial product. A comparison of performance on the standard CogState tests with the GMLT will help to elucidate unique aspects of executive function assessed by the GMLT and specific outcome metrics that are most useful in assessing this cognitive domain. Finally, knowledge regarding the construct validity of the GMLT in assessing processing speed, working memory, and several aspects of

executive function in healthy individuals is needed to both understand how healthy individuals perform on the test and to determine the factor structure of the outcome measures produced by this instrument.

This study was comprised of two sub-studies. Study 1 examined the convergent validity of the GMLT in assessing processing speed and working memory, and the divergent validity of the GMLT in assessing aspects of executive function in a sample of healthy adults. Specifically, performance on GMLT outcome measures was compared with performance on tests of psychomotor speed, working memory, and learning from the CogState (CGS; <http://www.cogstate.com/>) computerized cognitive test battery. A secondary aim of Study 1 was to determine the factor structure of the GMLT using exploratory factor analysis, and evaluate the validity of resultant factor(s) by examining their association with individual outcome measures and factor score(s) on the CogState tests. An examination of the correlations between performance on tests of general cognitive function and the GMLT using factor analysis should provide evidence for the extent to which executive function measured by the GMLT is independent of general attentional, psychomotor, and working memory abilities as assessed by the CogState tests.

Study 2 evaluated the stability of the GMLT factor structure over repeated assessments in a sample of 256 patients undergoing parathyroidectomies or thyroidectomies. This convenience sample was chosen because it allowed us to examine whether the factor structure of the GMLT determined in healthy adults in Study 1 is replicable in a clinical population in which the GMLT had previously been used to assess baseline cognitive impairment and post-parathyroidectomy cognitive improvement (Roman et al., 2005). Because this study evaluated GMLT performance up to 6 months post-surgery and contained many of the limitations inherent in clinical trials (e.g., loss to follow up), it provides a conservative test of the stability of the GMLT factor structure in a clinical population.

1. Methods

1.1. Study 1

1.1.1. Participants

A convenience sample of 49 medically, neurologically, and psychiatrically healthy Caucasian male ($N=43$; 87.8%) and female ($N=6$; 12.2%) adults from Austin, Texas, ranging in age from 19 to 53 ($M=32.15$, $S.D.=9.83$), and with a mean of 14 years of education ($S.D.=3.3$ years), was used for this study. These individuals were participating in a randomized, double-blind, placebo-controlled, parallel-group study that compared the sensitivity of a variety of cognitive tests and measures in assessing alprazolam and gabapentin dose-dependent effects (Snyder, Werth, et al., 2005). Baseline data from Day 1 of this study, prior to any drug administration, were used to examine the convergent validity of the GMLT. Three practice sessions were administered prior to baseline test administration in order to familiarize participants with all tasks. This study was approved by the Western Institutional Review Board in 2001, and all participants who met inclusion criteria provided informed consent prior to engaging in experimental procedures.

1.1.2. Neuropsychological measures

1.1.2.1. Groton Maze Learning Test (GMLT). The GMLT is computer-based neuropsychological measure of spatial working memory, learning efficiency, and error monitoring (Pietrzak et al., 2007; Snyder, Bednar, et al., 2005; Snyder, Werth, et al., 2005), and it is commercially available from CogState, Ltd. for academic and clinical use. It consists of a 10×10 grid of grey tiles presented on a computer touch-screen. To complete the maze, the participant has to follow a hidden pathway (28 moves, 11 turns) through the grid from the top left corner to a flag in the bottom right corner. Message bars at the top and bottom of the screen inform the participant whether a move is correct. If the move is correct, the participant is prompted to “Go On” by the message bar and a musical tone. If the move is incorrect, it is recorded as a “legal error” and the participant is instructed to go back to the last correct location (‘tile’) and to try moving in a different direction. If two incorrect responses are made in a row, the second consecutive error is labeled a “perseverative error” and the participant is again instructed to move back to the previous correct tile and to try a new way. If the participant fails to return to the last correct square after making two successive wrong moves (i.e., they make three consecutive incorrect moves), the third error is labeled a “rule-break error,” and the tile that corresponds to the last correct move begins to flash. The participant is then asked to touch the flashing tile and then try to continue through the maze. The trial ends once the participant reaches the flag in the bottom right corner of the grid of tiles.

Each participant completed five successive learning trials. Each trial was timed (in ms) and timing began automatically when the participant made his or her first move on each learning trial. In its entirety, the task took each participant 10–15 min to complete. A description of primary and secondary measures of the GMLT is shown in Table 1.

The GMLT also contains a brief subtest called the “timed chase test,” which assesses simple visuomotor processing speed. In this subtest, the participant is asked to move as rapidly as possible, one tile at a time, through the same 10×10 grid of grey tiles, while following a colored tile that keeps moving (essentially, a game of “follow the leader”). After all participants first complete an untimed practice run to learn the task, the number of correct moves made during a subsequent timed trial (30 s) is used as an index of visuomotor speed. This “matched” measure has the advantage of being administered with the exact same stimulus field, apparatus, and input device as the actual maze learning test, with the only differences being the absence of any requirement to learn a new hidden maze path and the request made to participants to make as many correct moves as possible within the 30 s time limit. It may also be entered as a statistical covariate to control for the potentially confounding influence of processing speed on spatial working memory and error monitoring, particularly in clinical populations with known impairment in motor function and processing speed.

1.1.2.2. CogState (CGS) measures. Details regarding the CGS test battery are available elsewhere (Collie, Maruff, Darby, & McStephen, 2003; Collie, Maruff, McStephen, & Darby, 2003; Darby et al., 2002; Maruff et al., 2006). Participants were required to respond with three keys on each subtest by using the “D” or “K” or “spacebar.” An error beep sounded when a participant pressed an incorrect key at any time. Dependent variables for each task included (1) accuracy of performance [the number of true positive and true negative responses made expressed as a percentage of the total number of trials]; (2) speed of performance [mean reaction time for responses on correct trials, initially recorded in milliseconds]; and (3) consistency of performance [the standard deviation of response times on correct trials].

Simple reaction time test (SRT): The SRT is a simple test of psychomotor speed. The participant is shown a single card at the center of the display. The instruction is given by the “on-screen” helper to “Press the spacebar as soon as the card turns over.” The participant then presses the spacebar as soon as the playing card turns face-up. Participants perform 18 repetitions on the first trial and 15 repetitions on the two subsequent trials.

Choice reaction time task (ChRT): The ChRT assesses psychomotor speed and visual attention. A white card with either a black or red square in the middle is presented on the screen. The participant is instructed to “Press K if the card is red and D if it is black”. Once the participant demonstrates that they know what to do, the cards revert to conventional playing cards and the participant now responds to the color of the suit of the card. The participant performs 17 repetitions.

Congruent reaction time task (CgRT): The CgRT assesses psychomotor speed and visual attention. Two white cards, aligned vertically, with either a black or red square are presented in the middle of the screen. The participant is given the instruction “Press K if the two cards are the same color and D if they are different.” Once the participant demonstrates they know what to do, the cards revert to conventional playing cards and the participant now responds as to whether the color of the suit of both cards is the same or different. The participant performs 17 repetitions.

One-back memory task (OBK): The OBK assesses working memory. In this task, two playing cards are presented in the middle of the screen, one face up and the other face down. The participant is given the instruction: “Press K if the card is the same as the one before. Press D if it is different.” The first answer is always no, as no prior card has been shown. After the participant presses the “D” key, the card flips over and a new card appears face-up. If this card is the same as the previous card then the participant presses “K”, if not they press “D”. When correct responses are recorded, the cards move to a single pack. The participant performs 17 repetitions.

Monitoring task (Mon): The Mon task assesses visual attention. Five cards are presented face up in the middle of the screen. White lines are marked above and below these cards. The five cards then begin to jiggle in either an upwards or downwards direction pseudo-randomly (predominant direction also is coded randomly in the computer program). The direction of each card is independent of each other card. The participant is given the instruction: “Press the space bar as soon as a card touches a white line.” When a card crosses a white line and participant presses the “K” key, it returns to the middle of the screen and starts to jiggle again. The participant performs this task until 17 cards have touched one of the white lines (i.e., 17 repetitions).

Matching task (MATCH): The MATCH task assesses working memory. Six pairs of cards (each pair aligned vertically) are presented in a display at the top of the computer screen. A single pair of cards (also aligned vertically) is presented at the bottom of the screen. Cards will always be paired with the same card if any one card is part of a pair

in the legend. The participant is instructed to “Press K if the two cards match a pair in the legend or press D if they do not.” As soon as the participant presses the “K” or “D” key a new pair of cards is shown at the bottom of the screen. A total of 24 repetitions are performed. Participants should try to develop a strategy to remember at least one of each pair of cards to prepare for the next task.

Incidental learning task (INCID): The INCID task assesses short-term memory. Once the participant has correctly identified a card pair at the bottom of the screen four times, all of the card pairs at the top of the screen turn face down. The participant must then remember the six card pairs that were at the top of the display. As each new card pair is presented at the bottom of the screen, the participant must determine if this pair is the same as a pair that was shown previously at the top of the screen, and now held in their memory. A total of 12 repetitions are performed.

Paired associate learning task (LEARN): The LEARN task assesses visual memory. Five card pairs (aligned vertically) are presented at the top of the screen. A single card pair (aligned vertically) is also presented at the bottom of the screen. The participant is instructed to “Press K if the two cards match a pair in the legend in any order or press D if they do not.” As soon as the participant presses the “K” or “D” key a new pair of cards is shown at the bottom of the screen. If the single pair of cards matches one of the pairs of cards in the display at the top of the screen, the pair at the top of the screen turns face down. This continues until all of the card pairs at the top of the screen are turned face down with the exception of the central pair which remains face-up as a control task. The participant must remember all the card pairs at the top of the screen, and determine whether or not the card pair at the bottom of the screen matches these remembered pairs. Cards in the single presented pair do not have to be in the same order or paired with the exact card of the pair. If the participant answers incorrectly, a buzzer is sounded and a visual instruction is activated to indicate that they were incorrect. They can also use this information in order to learn the pairs that were presented at the top of the display. Participants are presented with each correct card pair five times. Interspersed throughout these are card pairs that do not match any of the card pairs at the top of the display. Each of these incorrect pairs appears only once in the test. The participant performs 50 repetitions altogether.

1.1.3. Statistical analysis

Data that were non-normally distributed were transformed using logarithmic base 10 transformations. Pearson correlation coefficients were used to examine relationships between GMLT and CGS outcome measures and factor scores. Following the recommendations of MacCallum, Widaman, Zhang, and Hong (1999), exploratory factor analyses were used to determine the factor structure of the GMLT and CGS measures. Initially, the GMLT measures total time, total moves, maze efficiency index, legal errors, perseverative errors, and rule-break errors were entered into an exploratory factor analysis. In a subsequent exploratory factor analysis, total moves and total time were excluded because legal, perseverative, and rule-break errors accounted for excess moves on the GMLT, and because total time and maze efficiency index scores were highly correlated. All CGS measures were entered into an exploratory factor analysis. Incidental learning was excluded from this analysis because it did not load on the single factor that emerged. Eigenvalues meeting Kaiser’s criterion of >1 were used to determine the appropriate number of factors.

Table 2 shows GMLT measures, the cognitive constructs that they purportedly measure, and comparator measures from the CGS test battery that we expected to be associated with them.

1.2. Study 2

1.2.1. Participants

Data for this exploratory factor analytic study were drawn from a prospective case–control study examining the effect of parathyroidectomy on neurocognitive function in patients with primary hyperparathyroidism at the Yale University Medical Center. A total of 156 consecutive eligible patients with primary hyperparathyroidism undergoing parathyroidectomy and 98 consecutive eligible patients with benign thyroid disease undergoing thyroidectomy participated in this study. All eligible consenting English-speaking patients were included from June 2004 until February 2007. Exclusion criteria were age younger than 18 years; inability to read and write at an 8th grade level; abnormal renal function, as documented by serum creatinine concentration; hyper- or hypothyroidism as documented by serum thyroid-stimulating hormone; a recent diagnosis of cancer; pregnancy; frank psychosis; severe dementia; and active substance abuse. Patients were assessed 2 weeks prior to surgery (baseline), and 2 weeks, 2 months, and 6 months

Table 2

Groton Maze Learning Test measures, cognitive constructs, and comparator measures from CogState battery

GMLT measures	Cognitive construct	Comparator measures from CogState battery
Timed chase test	Psychomotor/processing speed	Simple reaction time Choice reaction time Congruent reaction time Monitoring task
Maze efficiency index	Spatial memory efficiency	Simple reaction time Choice reaction time Congruent reaction time Monitoring task Matching task One-back test Associate learning test Incidental learning test
Legal errors	Spatial working memory	Monitoring task Matching task One-back test Associate learning test Incidental learning test
Perseverative errors	Executive function (error monitoring)	Monitoring
Rule-break errors	Executive function (error monitoring)	Monitoring
Total errors	Executive function (error monitoring)	Monitoring

after surgery. The study was approved by the Human Investigations Committee of Yale University and all participants provided written informed consent.

1.2.2. Demographic and neuropsychological measures

Demographic data (e.g., age, sex, race) and clinical information, including comorbidities, history of endocrinopathies, and prior psychiatric illnesses were obtained during the presurgical evaluation. Trained psychometrists administered the GMLT during the presurgical and at three postbaseline follow-up examinations. The same investigators performed all data collection at each evaluation.

1.2.3. Statistical analysis

Exploratory factor analysis was used to examine the factor structure of the GMLT at baseline, and at 2 weeks, 2 months, and 6 months post-surgery. Eigenvalues meeting Kaiser's criterion of >1 were used to determine the appropriate number of factors at each assessment.

2. Results

2.1. Study 1

2.1.1. Convergent validity of the Groton Maze Learning Test

Table 3 shows a correlation matrix of scores on the GMLT and CogState tests. Significant correlations between GMLT outcome measures and comparator CGS test outcome measures ranged in magnitude from 0.29 to 0.62 (medium to large effect size; Cohen, 1988). GMLT-timed chase test scores correlated positively with CGS-incidental learning, and negatively with CGS-simple reaction time, choice reaction time, congruent reaction time, monitoring, one-back test, matching, and associate learning. GMLT-maze efficiency index scores correlated negatively with CGS measures of simple reaction time, choice reaction time, congruent reaction time, monitoring, one-back test, matching, and associate learning. GMLT-legal errors correlated positively with CGS measures of simple reaction time, congruent reaction time, and monitoring. GMLT-perseverative errors correlated positively with CGS-congruent reaction time. GMLT-rule-break errors correlated positively with CGS-choice reaction time, congruent reaction time, and matching

Table 3
Correlation matrix of Groton Maze Learning Test and CogState measures

	GMLT-timed chase test	GMLT-maze efficiency index	GMLT-legal errors	GMLT-perseverative errors	GMLT-rule-break errors	GMLT-total errors	GMLT-learning efficiency factor	GMLT-error monitoring factor
CogState comparator measures								
Simple reaction time	−0.511***	−0.618***	0.312*	0.193	0.252	0.321*	−0.606***	0.158
Choice reaction time	−0.561***	−0.476***	0.156	0.180	0.296*	0.187	−0.525***	0.137
Congruent reaction time	−0.625***	−0.512***	0.334*	0.350*	0.412**	0.377**	−0.569***	0.311*
Monitoring	−0.593***	−0.555***	0.401**	0.256	0.211	0.404**	−0.620***	0.190
One-back	−0.514***	−0.318*	0.012	0.083	0.183	0.038	−0.413**	0.021
Matching	−0.517***	−0.423**	0.247	0.102	0.291*	0.253	−0.509***	0.127
Incidental learning	0.295*	−0.103	0.179	0.191	0.067	0.190	0.105	0.209
Associate learning	−0.504***	−0.466***	−0.062	−0.119	0.157	−0.061	−0.520***	−0.130
Learning efficiency factor	−0.698***	0.616***	−0.252	−0.189	−0.330*	−0.274	0.687***	−0.147

Note—GMLT: Groton Maze Learning Test; CGS: CogState.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

Table 4
Factor loadings for Groton Maze Learning Test measures

	Factor 1: Error monitoring	Factor 2: Learning efficiency
GMLT outcome measures		
Timed chase test	−0.050	0.813
Maze efficiency index	−0.179	0.882
Legal errors	0.635	−0.517
Perseverative errors	0.925	0.028
Rule-break errors	0.842	−0.205

Note: Bold values indicate high loadings on factor.

tests. GMLT-total errors correlated positively with CGS measures of simple reaction time, congruent reaction time, and monitoring.

2.1.2. Exploratory factor analysis of GMLT outcome measures

Exploratory factor analysis of GMLT measures suggested a two-factor solution. Correlation matrix eigenvalues were 2.56 and 1.19 before rotation (75.0% of total variance explained). The first factor, termed “error monitoring,” consisted of positive loadings of legal errors, perseverative errors, and rule-break errors; the inverse of these factor scores was termed “error monitoring.” The second factor, termed “learning efficiency,” consisted of positive loadings of timed chase test and maze efficiency index scores, and negative loadings of legal errors. Both varimax (uncorrelated) and promax (correlated) rotation loadings suggested the same factor solution. Table 4 shows uncorrelated varimax factor loadings.

2.1.3. Exploratory factor analysis of CogState tests

Exploratory factor analysis of scores on the CGS tests suggested a one-factor solution. The correlation matrix eigenvalue of this factor was 4.27 before rotation (61.0% of total variance explained). The inverse of this factor, termed “learning efficiency,” consisted of positive loadings of all CGS tests, except incidental learning, which did not load on this factor. Both varimax (uncorrelated) and promax (correlated) rotation loadings suggested the same factor solution. Varimax factor loadings ranged from 0.745 to 0.838.

As shown in Table 3, the GMLT-learning efficiency factor correlated positively with the CGS-learning efficiency factor, and negatively with all CGS tests except incidental learning. The GMLT-error monitoring factor was not correlated with the CGS-learning efficiency factor, but was correlated positively with scores on the CGS-congruent reaction time test. Fig. 1 shows a scatterplot of scores on the GMLT and CogState learning efficiency factors.

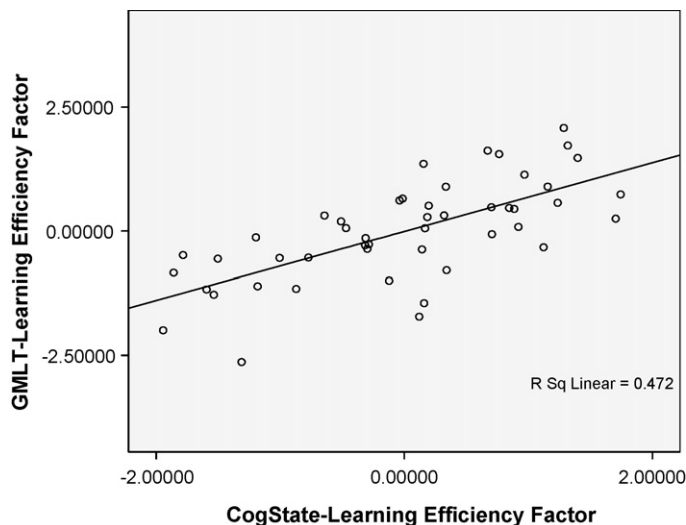


Fig. 1. Scatterplot of CogState × GMLT learning efficiency factor scores.

Table 5

Factor loadings for Groton Maze Learning Test measures at baseline, and 2 weeks, 2 months, and 6 months post-surgery

	Factor 1: Error monitoring	Factor 2: Learning efficiency
GMLT outcome measures		
Baseline (<i>N</i> = 256)		
Timed chase test	−0.014	0.897
Maze efficiency index	−0.271	0.872
Legal errors	0.782	−0.437
Perseverative errors	0.980	−0.098
Rule-break errors	0.974	−0.046
2 weeks post-surgery (<i>N</i> = 179)		
Timed chase test	−0.026	0.884
Maze efficiency index	−0.259	0.853
Legal errors	0.702	−0.494
Perseverative errors	0.969	−0.131
Rule-break errors	0.962	−0.062
2 months post-surgery (<i>N</i> = 122)		
Timed chase test	−0.159	0.899
Maze efficiency index	−0.296	0.843
Legal errors	0.756	−0.419
Perseverative errors	0.952	−0.199
Rule-break errors	0.952	−0.175
6 months post-surgery (<i>N</i> = 97)		
Timed chase test	−0.103	0.878
Maze efficiency index	−0.186	0.866
Legal errors	0.713	−0.345
Perseverative errors	0.940	−0.096
Rule-break errors	0.933	−0.079

Note: Bold values indicate high loadings on factor.

The CGS-learning efficiency factor correlated positively with GMLT-maze efficiency index scores, and negatively with GMLT-timed chase test moves and rule-break errors.

2.2. Study 2

Exploratory factor analysis of GMLT measures suggested a two-factor solution at each point of assessment. At baseline, correlation matrix eigenvalues were 3.02 and 1.34 before rotation (87.2% of total variance explained). At 2 weeks post-surgery, these values were 2.94 and 1.26, respectively (84.0% of total variance explained). At 2 months post-surgery, these values were 3.26 and 1.15, respectively (85.3% of total variance explained). At 6 months post-surgery, these values were 2.72 and 1.24, respectively (79.2% of total variance explained). As in Study 1, the first factor, termed “error monitoring,” consisted of positive loadings of legal errors, perseverative errors, and rule-break errors; the inverse of these factor scores was termed “error monitoring.” The second factor, termed “learning efficiency,” consisted of positive loadings of timed chase test and maze efficiency index scores, and negative loadings of legal errors. Both varimax (uncorrelated) and promax (correlated) rotation loadings suggested the same factor solution. Table 5 shows uncorrelated varimax factor loadings at each point of assessment.

The factor structure remained the same over repeated assessment even when analyzed separately for patients undergoing parathyroidectomies and patients undergoing thyroidectomies.

3. Discussion

This study examined the construct validity of the GMLT with various measures from the CogState neuropsychological test battery. Several GMLT measures correlated moderately-to-strongly with CogState measures (r 's = 0.29–0.62), with the timed chase test and maze efficiency index measures being most consistently correlated with all CGS measures

except incidental learning. Conversely, GMLT error measures were infrequently associated with CGS measures. The GMLT-learning efficiency factor correlated strongly with the CGS learning efficiency ($r=0.69$), but the GMLT-error monitoring factor did not ($r=-0.15$). This factor structure was stable over repeated assessments in a separate sample of patients undergoing parathyroidectomy or thyroidectomy. These results and recommendations for further research are discussed.

GMLT-timed chase test and maze efficiency scores correlated moderately-to-strongly with CGS measures of reaction time, working memory, and learning. This finding suggests that the GMLT-timed chase test relies on reaction time and visual attention, and that it is related to measures of working memory and learning that require rapid responding. Because the GMLT and CGS measures are timed, visuomotor processing speed clearly plays a role in how well an individual performs on these tests. The maze efficiency index, one of the primary outcome measures of the GMLT, also appears to rely on processing speed, as individuals with low scores on this measure (i.e., those who were less efficient in learning a hidden maze path) had longer reaction times and performed worse on CGS measures of visual attention, working memory, and learning. Further, exploratory factor analysis of GMLT outcome measures showed that timed chase test and maze efficiency scores loaded on the same factor, thereby suggesting that visuomotor speed is related to maze learning efficiency. Taken together, these results demonstrate the validity of the GMLT-timed chase test as a measure of simple visuomotor processing speed, and the GMLT-maze efficiency index as a measure of processing speed, visual attention, and working memory for spatial information.

GMLT-rule-break and total errors correlated moderately with CGS measures of reaction time, matching, and monitoring. This pattern of correlations suggests that rule-break and total error scores on the GMLT may rely on psychomotor speed, attention, and working memory. Monitoring of performance and response inhibition (i.e., avoiding making rule-break errors) on the GMLT is required for optimal performance on the test. Thus, increased rule-break and total errors on the GMLT appear to be related to slower psychomotor speed, and poorer attention and working memory.

Exploratory factor analysis of GMLT outcome measures suggested a two-factor solution—error monitoring and learning efficiency. This finding is consistent with a growing number of factor analytic studies of the GMLT in both normative and clinical samples. Results of these studies indicate that the GMLT assesses how well an individual monitors their performance while learning a hidden maze, and how efficiently they are able to learn a hidden maze. The finding that the GMLT-learning efficiency factor correlated strongly ($r=0.69$) with the CGS-learning efficiency factor, but the GMLT-error monitoring factor did not ($r=-0.15$) demonstrates the convergent validity of the GMLT-learning efficiency factor, and the divergent validity of the GMLT-error monitoring factor. The two-solution factor structure of error monitoring and learning efficiency was also found in a sample of patients undergoing parathyroidectomy or thyroidectomy who were assessed presurgically, and at three postbaseline follow-up periods extending over a 6-month period. These findings provide the first demonstration of the stability of the GMLT factor structure over repeated assessments.

The GMLT-error monitoring factor was not associated with the CGS-learning efficiency factor. This finding supports the divergent validity of the GMLT, which assesses a novel aspect of executive function not assessed by the CGS measures included in this study: error monitoring. It also supports the inclusion of the GMLT in the existing CogState test battery. Previous studies have found that error scores on the GMLT are correlated with performance on other well-known tests of processing speed, attention, and planning (e.g., Pietrzak et al., 2007; Pietrzak, Sprague, & Snyder, under review). In this study, the GMLT-error monitoring factor correlated strongly with the GMLT measure of total errors ($r=0.76$). Thus, in studies utilizing within-subject designs, such as in clinical trials that require multiple assessments, total errors on the GMLT would likely serve as the most useful measure of error monitoring ability. Indeed, the current CogState version of the GMLT utilizes total errors and rule-break errors, the latter being the sum of perseverative and rule-break errors, as the main outcome measures. Additional research is needed to evaluate the sensitivity of the combined perseverative and rule-break error measure in detecting treatment-related cognitive change.

Methodological limitations of this study must be noted. First, the sample in Study 1 was relatively small, consisted predominantly of men, and was ethnically and geographically homogeneous. Additional studies using larger and more heterogeneous samples are needed to confirm these results. Second, this study focused on the convergent, but not divergent or discriminant validity of the GMLT. More research is needed to tease apart the specific aspects of cognitive function assessed by the GMLT by comparing GMLT performance to a wider range of cognitive tasks, including those that measure aspects of cognition unrelated to those purportedly assessed by the GMLT. Third, the sample size-to-variable ratio for exploratory factor analyses was relatively small according to some sources (Tabachnick & Fidell, 2007), suggesting that these results require replication in larger samples. Finally, familiarity with video game-like tasks

was not assessed, and individuals with increased familiarity with such tasks may perform better on GMLT and CGS tasks. Future research using computerized cognitive tasks should examine whether familiarity with video game-like tasks influences neuropsychological test performance (e.g., Driscoll, Hamilton, Yeo, Brooks, & Sutherland, 2005).

Despite these limitations, results of this study provide a preliminary demonstration of the construct validity of the GMLT in assessing visuomotor processing speed, spatial working memory, and error monitoring ability. These findings support the continued use of the GMLT in clinical research, as it is currently being applied by pharmaceutical companies and academic research groups in the United States, England, France, Netherlands, Germany, Singapore, Japan, and Australia. Although the clinical diagnostic utility of this test has been explored recently in a variety of patient populations, ranging from children with ADHD, to patients with schizophrenia and mild cognitive impairment, further research is still required to examine the convergent and divergent validity of the GMLT relative to other neuropsychological tests or batteries (Delis–Kaplan Executive Function Scale; Delis, Kaplan, & Kramer, 2001), particularly those that assess various aspects of executive function (e.g., problem solving, planning and reasoning, cognitive flexibility) and to evaluate its construct validity and factor structure in various clinical populations.

Disclosure of interest

Dr. Maruff is Chief Scientific Officer and Dr. Snyder is a consultant for CogState, Ltd.

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